SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 - ASSESSMENT TOOL 2 – Coding Challenge

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO5 – Coding Challenge**	**Assessment:**	**Assessment Tool 2 – Coding Challenge**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO5 Statement:	To understand the concept of supervised and unsupervised methods in machine learning		
Student Name:	Kishore Kumar. P	**Reg. No.:**	192424108
Section / Batch:	2024	**Date:**	27/8/2026

Code of Conduct

I Kishore Kumar. P (192424108) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Kshore Kumar. P

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. Extend your CART implementation to support entropy/information gain as an alternative splitting criterion, controllable via a parameter. Run both on the same dataset and report where the resulting trees differ in structure.

Code:

```
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier, export_text
X, y = load_iris(return_X_y=True)

gini_tree = DecisionTreeClassifier(criterion="gini", random_state=42)
gini_tree.fit(X, y)

entropy_tree = DecisionTreeClassifier(criterion="entropy", random_state=42)
entropy_tree.fit(X, y)

print("TREE USING GINI INDEX (CART):")
print(export_text(gini_tree, feature_names=load_iris().feature_names))

print("\nTREE USING ENTROPY / INFORMATION GAIN:")
print(export_text(entropy_tree, feature_names=load_iris().feature_names))

gini_structure = export_text(gini_tree, feature_names=load_iris().feature_names)
entropy_structure = export_text(entropy_tree, feature_names=load_iris().feature_names)

print("\nCOMPARISON:")
if gini_structure == entropy_structure:
    print("Both trees have the same structure.")
else:
    print("The trees differ in structure.")
    print("Gini selects splits using impurity reduction.")
    print("Entropy selects splits using Information Gain.")
```

Output:

```
TREE USING GINI INDEX (CART):
|--- petal length (cm) <= 2.45
|   |--- class: 0
|--- petal length (cm) >  2.45
|   |--- petal width (cm) <= 1.75
|   |   |--- petal length (cm) <= 4.95
|   |   |   |--- petal width (cm) <= 1.65
|   |   |   |   |--- class: 1
|   |   |   |--- petal width (cm) >  1.65
|   |   |   |   |--- class: 2
|   |   |--- petal length (cm) >  4.95
|   |   |   |--- petal width (cm) <= 1.55
|   |   |   |   |--- class: 2
|   |   |   |--- petal width (cm) >  1.55
|   |   |   |   |--- petal length (cm) <= 5.45
|   |   |   |   |   |--- class: 1
|   |   |   |   |--- petal length (cm) >  5.45
|   |   |   |   |   |--- class: 2
|   |--- petal width (cm) >  1.75
|   |   |--- petal length (cm) <= 4.85
|   |   |   |--- sepal length (cm) <= 5.95
|   |   |   |   |--- class: 1
|   |   |   |--- sepal length (cm) >  5.95
|   |   |   |   |--- class: 2
|   |   |--- petal length (cm) >  4.85
|   |   |   |--- class: 2
```

```
TREE USING ENTROPY / INFORMATION GAIN:
|--- petal length (cm) <= 2.45
|   |--- class: 0
|--- petal length (cm) >  2.45
|   |--- petal width (cm) <= 1.75
|   |   |--- petal length (cm) <= 4.95
|   |   |   |--- petal width (cm) <= 1.65
|   |   |   |   |--- class: 1
|   |   |   |--- petal width (cm) >  1.65
|   |   |   |   |--- class: 2
|   |   |--- petal length (cm) >  4.95
|   |   |   |--- petal width (cm) <= 1.55
|   |   |   |   |--- class: 2
|   |   |   |--- petal width (cm) >  1.55
|   |   |   |   |--- petal length (cm) <= 5.45
|   |   |   |   |   |--- class: 1
|   |   |   |   |--- petal length (cm) >  5.45
|   |   |   |   |   |--- class: 2
|   |--- petal width (cm) >  1.75
|   |   |--- petal length (cm) <= 4.85
|   |   |   |--- sepal length (cm) <= 5.95
|   |   |   |   |--- class: 1
|   |   |   |--- sepal length (cm) >  5.95
|   |   |   |   |--- class: 2
|   |   |--- petal length (cm) >  4.85
|   |   |   |--- class: 2


COMPARISON:
Both trees have the same structure.
```


GPS Map Camera
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